

2023 NATIONAL YOUTH RISK BEHAVIOR SURVEY

Co-Occurring Adverse Experiences & Protective Factors in Adolescent Suicidality

A survey-weighted analysis of self-reported
adversity, protective behaviours, and suicide-risk
outcomes among U.S. high school students



Jacinta · Aliya · Destiny · Albert · Terence · Francis · Bernard



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01

Introduction

An introduction to our project,
key aims and objectives

Background



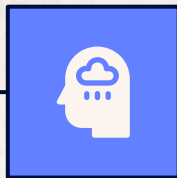
2021



Emergency

American Academy of Child and Adolescent Psychiatry declared a "national emergency in child and adolescent mental health" in the U.S.

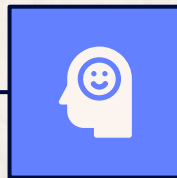
2022



Hearing

The U.S. Senate HELP Committee held a formal hearing, *"Why Are So Many American Youth in a Mental Health Crisis?"*

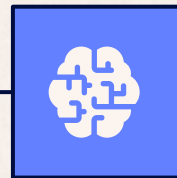
2023



Evidence

CDC's national survey quantifies the scale of the crisis among U.S. high schoolers

2026



Investigation

Our project examines which adverse and protective experiences co-occur with this risk

Background

1 in 5

seriously considered attempting
suicide in the past 12 months

39.7%

of students reported persistent
sadness or hopelessness
in the past 12 months

9.5%

attempted suicide one or more times
in the past 12 months

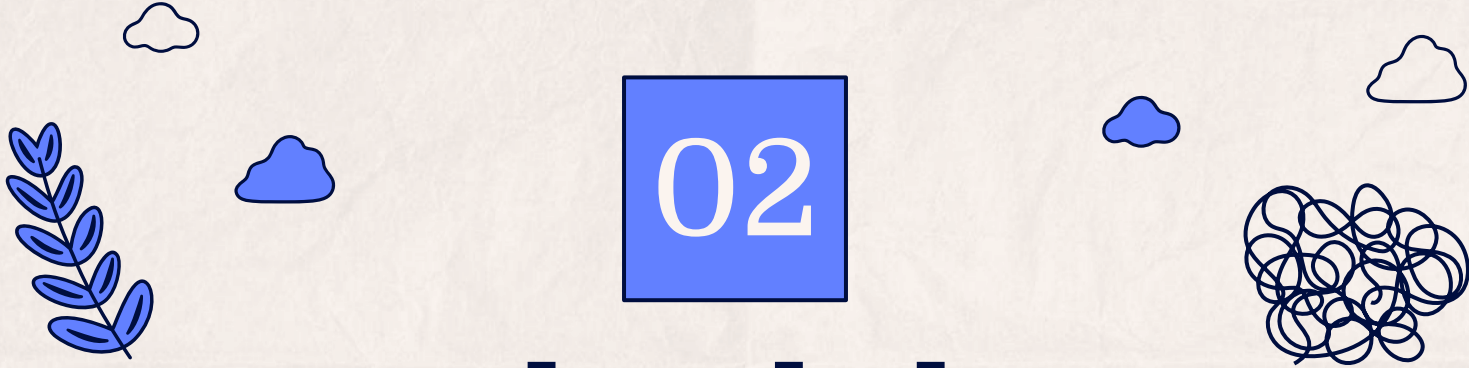




Learning More

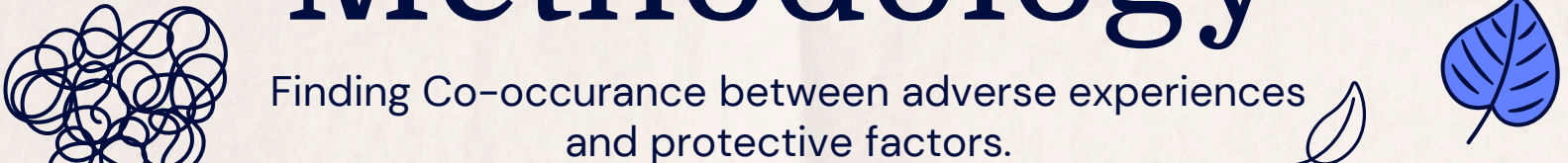
Suicide risk rarely comes from one thing.
This gap motivates our research question:

how do adverse and protective experiences
co-occur with adolescent suicidality, and
what does that mean for where prevention
efforts should focus?



02

Methodology



Finding Co-occurrence between adverse experiences
and protective factors.

Compare our independent analysis with current
research

Study design: 2023 National YRBS



What is it?

A nationally representative, biennial CDC survey of U.S. high school students (grades 9–12)

Anonymous, self-report questionnaire: every value in this dataset is what a student said, not something independently verified

20,103 usable student responses in the 2023 cycle measuring over 100 different variables.

Three-stage stratified cluster design:
strata → primary sampling units (PSUs, roughly clusters of schools) → classrooms



Why does this design matter?

Weight

Each respondent stands in for a number of real students nationally (not every student in the survey represents just one person)

stratum / psu

Students in the same school answer more alike than random students elsewhere.

Ignoring this understates uncertainty.

Our Variables

	Domain	Details	Variables
Jacinta	Demographics	Age, Ethnicity etc	Q1, Q2, Q3, raceeth
Aliya	Protective factors	Grades, Sleep, Basic Needs Met, Social life	QN85, QN87, QN99, QN103, QN104
Destiny	Adverse factors	Sexual dating and violence	QN19, QN20, QN21, QN88
Albert	Adverse factors	School and community	QN22, QN23, QN24, QN25, QN86
Terence	Adverse factors	Household dysfunction	QN89, QN90, QN91, QN100, QN101, QN102
Francis	Risk behaviours	Violence and mental distress	QN12, QN13, QN16, QN26, QN84
Bernard	Risk behaviours	Substance use	QN43, QN48, qntb4, qnillict, qnowt, qnobese

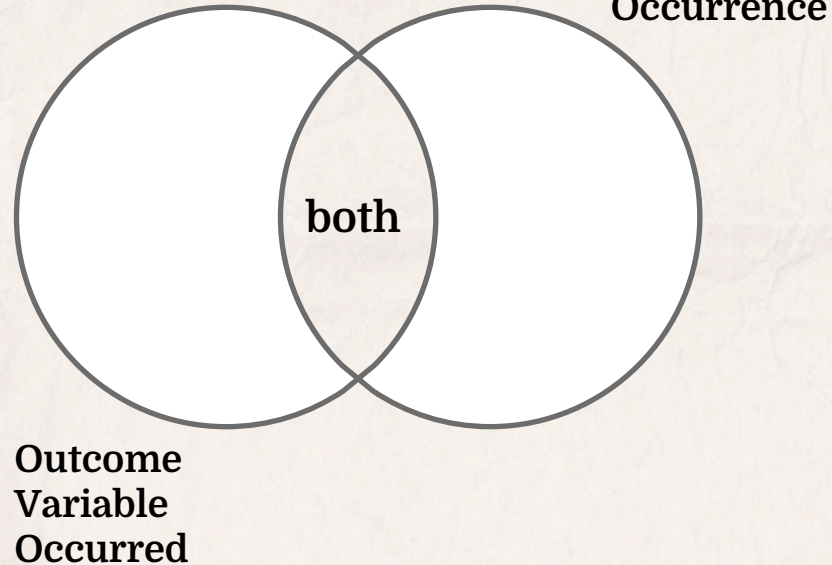
Variable Research, with cited Publications for the inclusion on each factor

Outcome Variables

"QN27", # seriously considered suicide (12mo)
"QN28", # made a suicide plan (12mo)
"QN29", # attempted suicide 1+ times (12mo)
"QN30" # attempt resulted in injury

p-values, why not?

All held significance, as such, results had no value.



Reproducibility via GitHub

Everything lives in one place

All scripts, data, documentation, and outputs are tracked in a single public repository to see the full history of the project

Every change is logged and attributed

21 commits across the project, each with a description of what changed and who changed it (from variable selection to charts)

Structured for reproducibility

Separate folders for data, scripts, output, markdown, and documents the pipeline from raw data to final figures is traceable end to end.

Analysis can be re-run from scratch

Anyone can rebuild our results because the R scripts and outputs are version-controlled together

Commits

History for BIP-Technical-Challenge-Collaboration / documents on [main](#)

Commits on Aug 17, 2026

Risk Behaviours

 FrancisAsane authored 3 days ago

Add files via upload

 ali-pixel545 authored 3 days ago

Commits on Aug 16, 2026

Add files via upload

 ajsaestheticjourneys authored 4 days ago

Commits on Aug 14, 2026

Updated to include QNLLICIT variable

 bernkyal authored last week

Add files via upload

 Terence03-art authored last week

Commits on Aug 12, 2026

Add files via upload

 ajsaestheticjourneys authored last week

Commits on Aug 9, 2026

Add files via upload

 cakemaster1017 authored 2 weeks ago

Add files via upload

 ali-pixel545 authored 2 weeks ago

Commits on Aug 5, 2026

Variable Research

 bernkyal authored 2 weeks ago

Collaboration via Google Drive & Teams

Shared with me > BIP Technical Challeng... > Recordings ▾

◆ Ask Gemini

Type ▾

People ▾

Modified ▾

Source ▾

Name ↓

Owner

More info (Option + →)

Date modified



Wednesday Meeting Team 04.-20260729_141217...



chukwuedojacinta

29 Jul



Team 04 Meeting.-20260802_192840-Meeting R...



chukwuedojacinta



Team 04 Meeting.-20260726_190616-Meeting Re...



chukwuedojacinta

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A survey-weighted analysis of self-reported adversity, protective behaviours, and suicide-risk outcomes among U.S. high school students

DI
Destiny Umayahwa Shu...

BK
Bernard Kyndel

AO
Aliya Umayahwa Shu...

FA
Francis Umayahwa Shu...

00:02:39

Jacinta · Aliya · Destiny · Albert · Terence · Francis · Bernard

HDR UK BIP Presentation Demo! - 20260820_19...



03

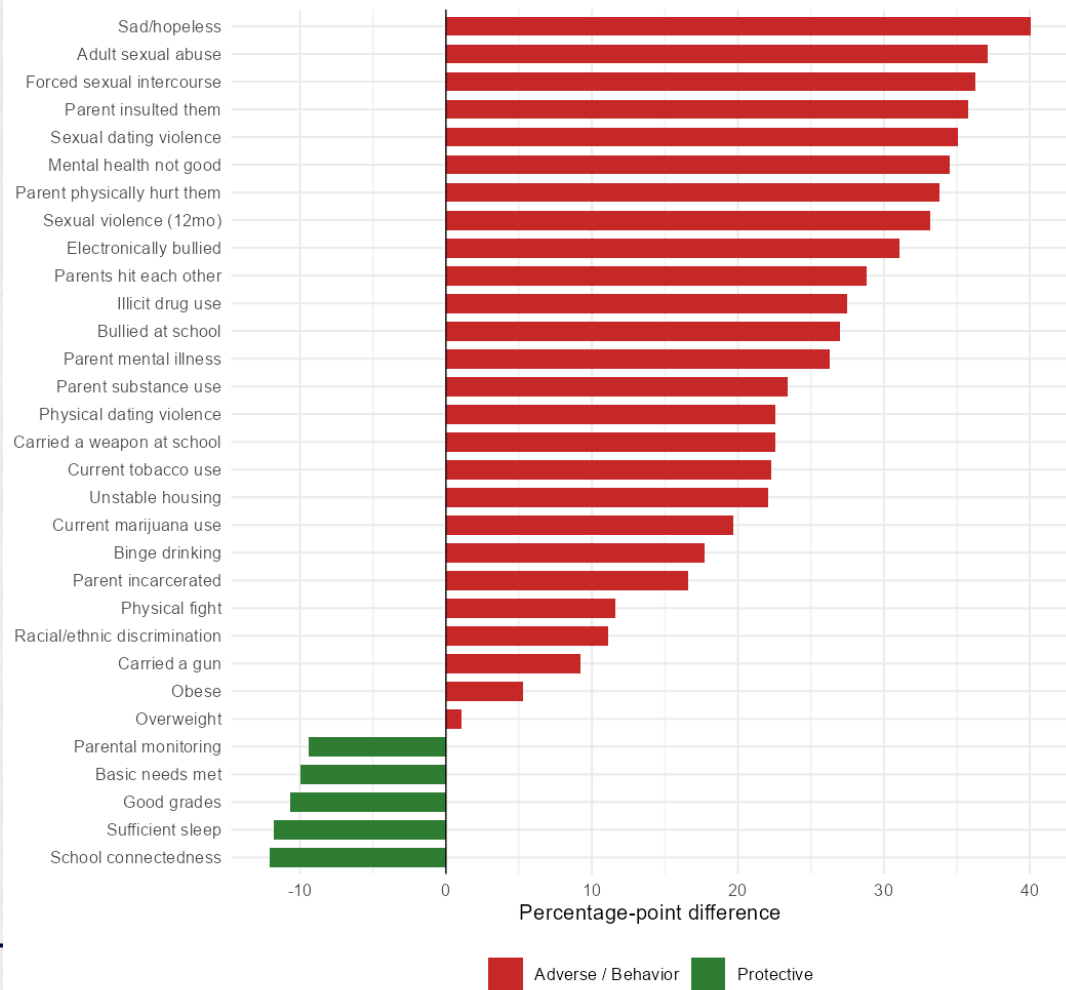
Key findings

A breakdown of key results
across the four suicide outcomes



Compared our outcome variables to
each of our selected variables

Considered suicide

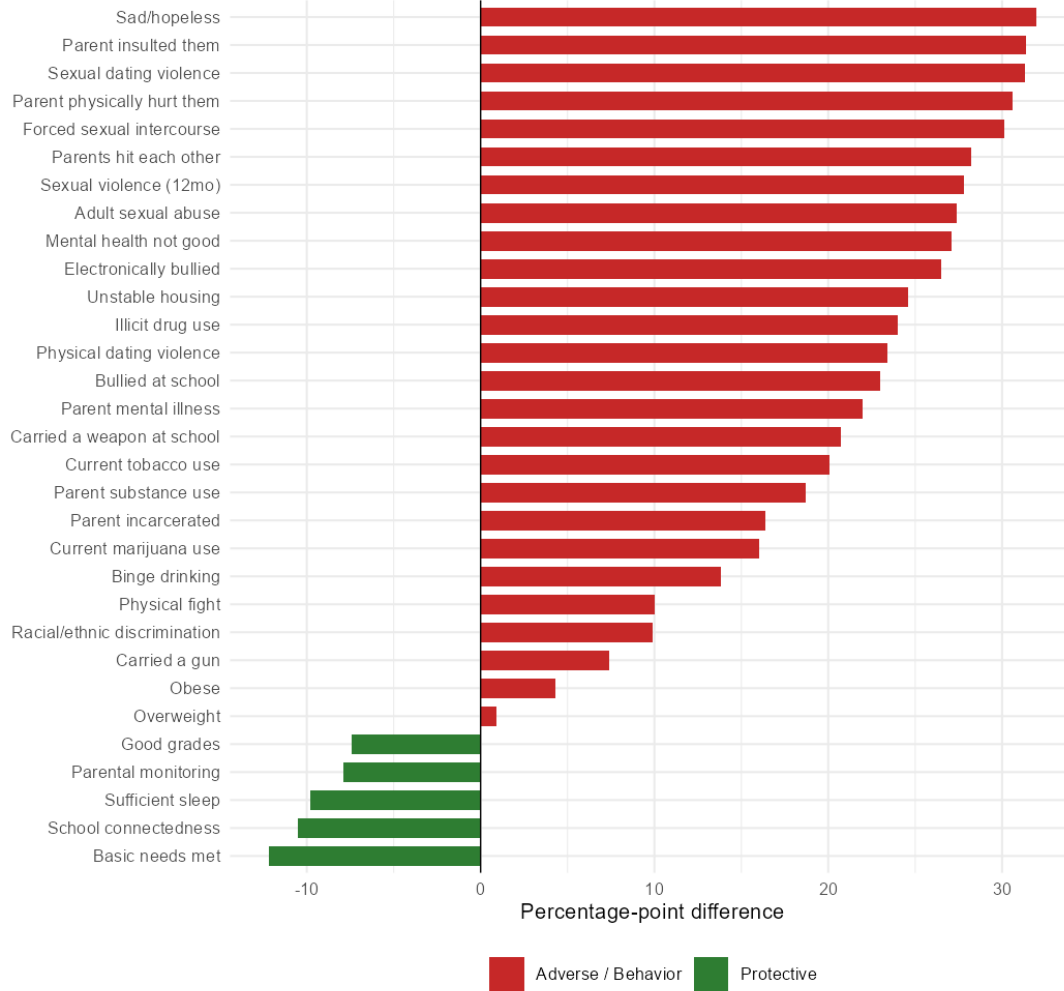


Cooccurrence

- Sad/Hopelessness
- Adult Sexual Abuse
- Forced Sexual Intercourse
- School connectedness
- Sleep
- Basics Needs Met



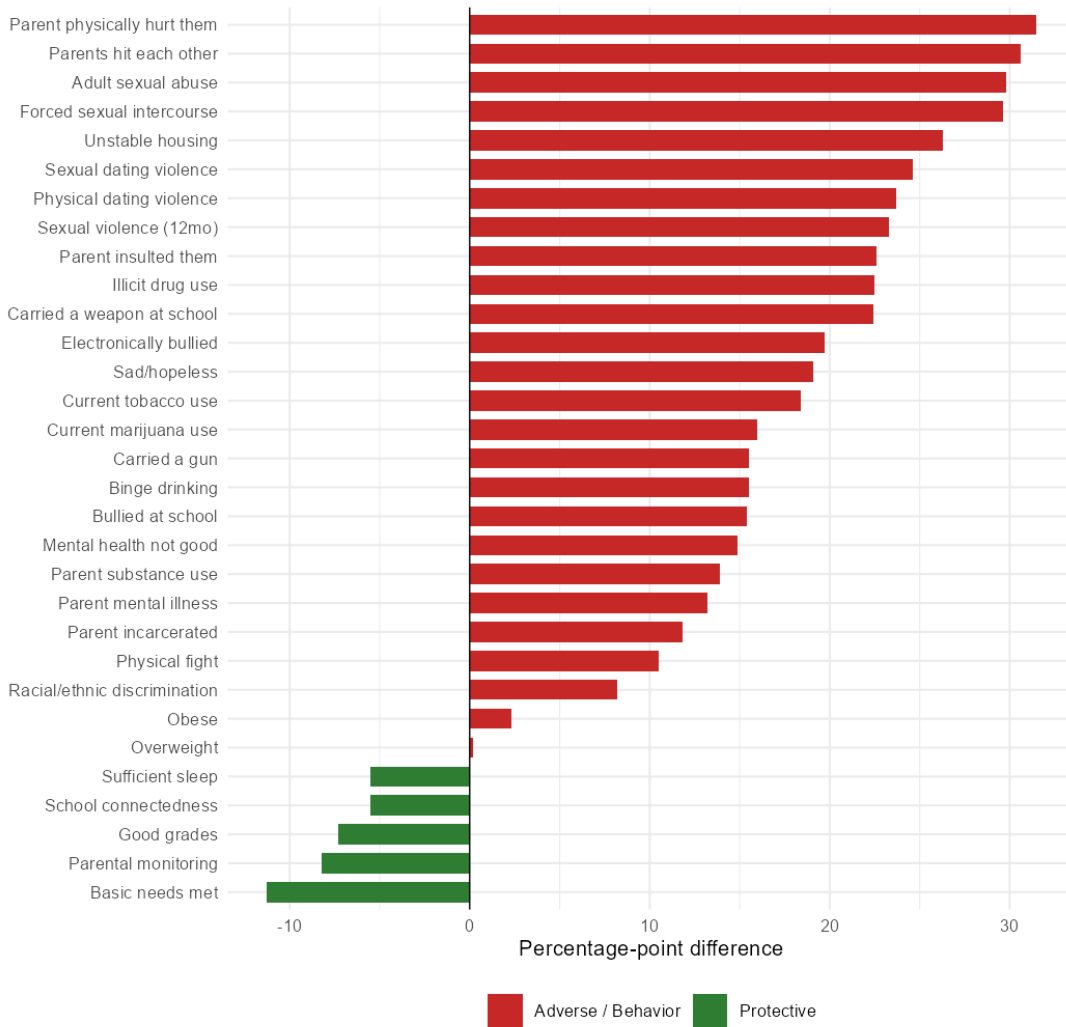
Made
a plan



Cooccurrence

- Sad/Hopelessness
 - Parents Insulted Them
 - Sexual Dating Violence
- ↑
- Basic Needs Met
 - School Connectedness
 - Sufficient Sleep
- ↓

Attempted

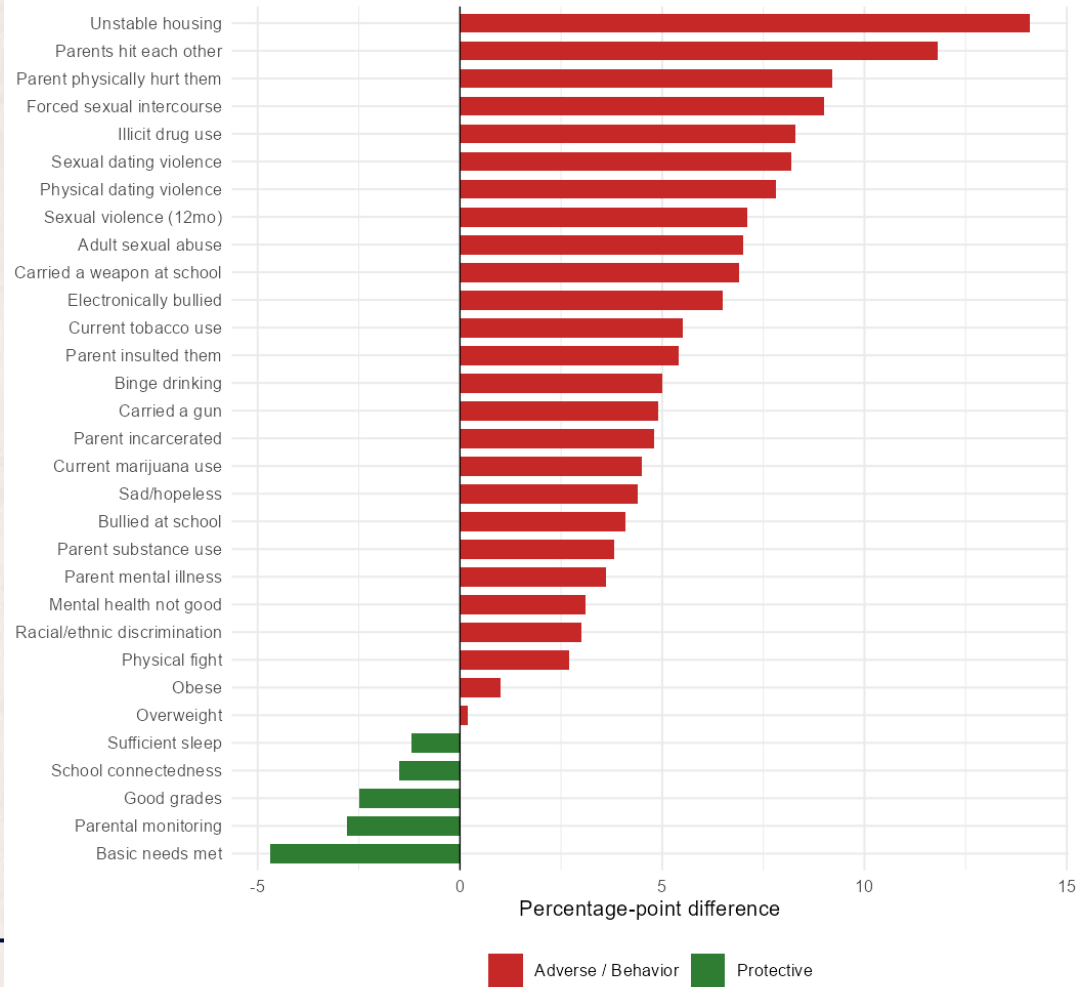


Cooccurrence

- Parent's physically hurt them
- Parents hitting each other
- Adult Sexual Abuse
- Basic Needs Met
- Parental monitoring
- Good Grades



Injurious attempt



Cooccurrence

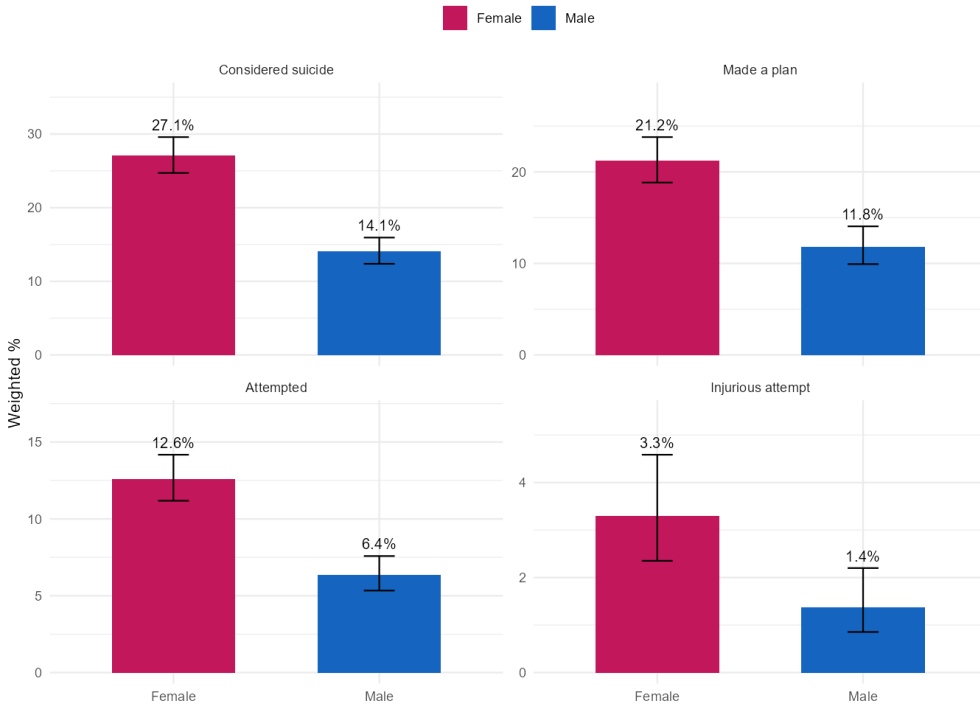
- Unstable Housing
- Parent's hitting each other
- Forced Sexual Intercourse
- Basic Needs Met
- Parental Monitoring
- Good Grades



Demographics (Jacinta's Variables)

Suicidality Outcomes by Sex

Weighted % with 95% CI, all four outcomes



Overall

Sex shows the strongest and most consistent demographic pattern in this dataset.

Female students report significantly higher prevalence than male students across all four outcomes

Composition

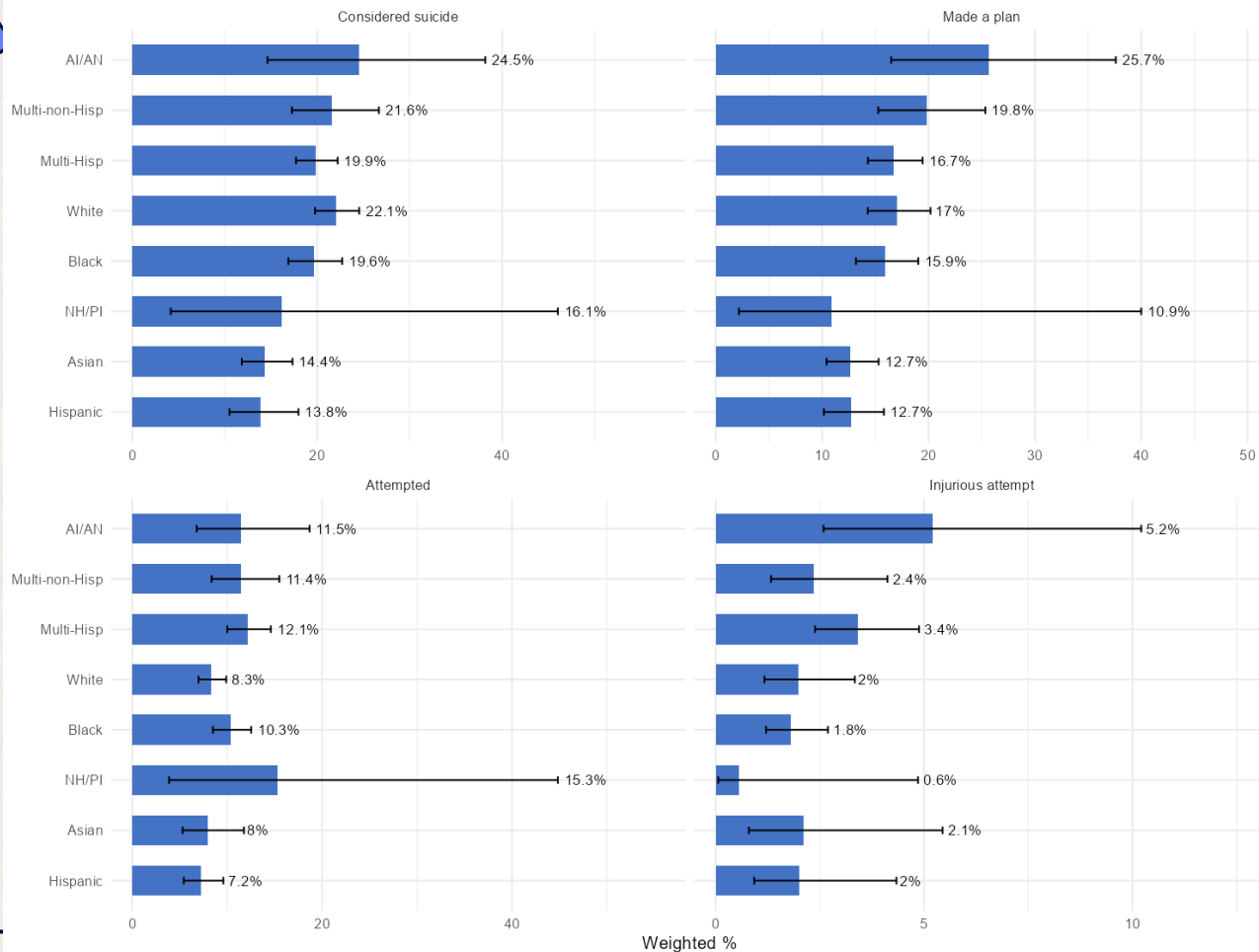
- White 48.1%
- Multi-Hispanic/Latino 19.8%
- Black 13.3%
- Hispanic 7.7%
- Multi-non-Hispanic 6.1%
- Asian 4.3%

Key findings

Race/ethnicity's relationship with suicidality is inconsistent across the four outcomes, statistically significant for considered suicide ($p=0.003$) and attempted suicide ($p=0.008$), but not significant for made a plan ($p=0.100$) or injurious attempt ($p=0.082$).

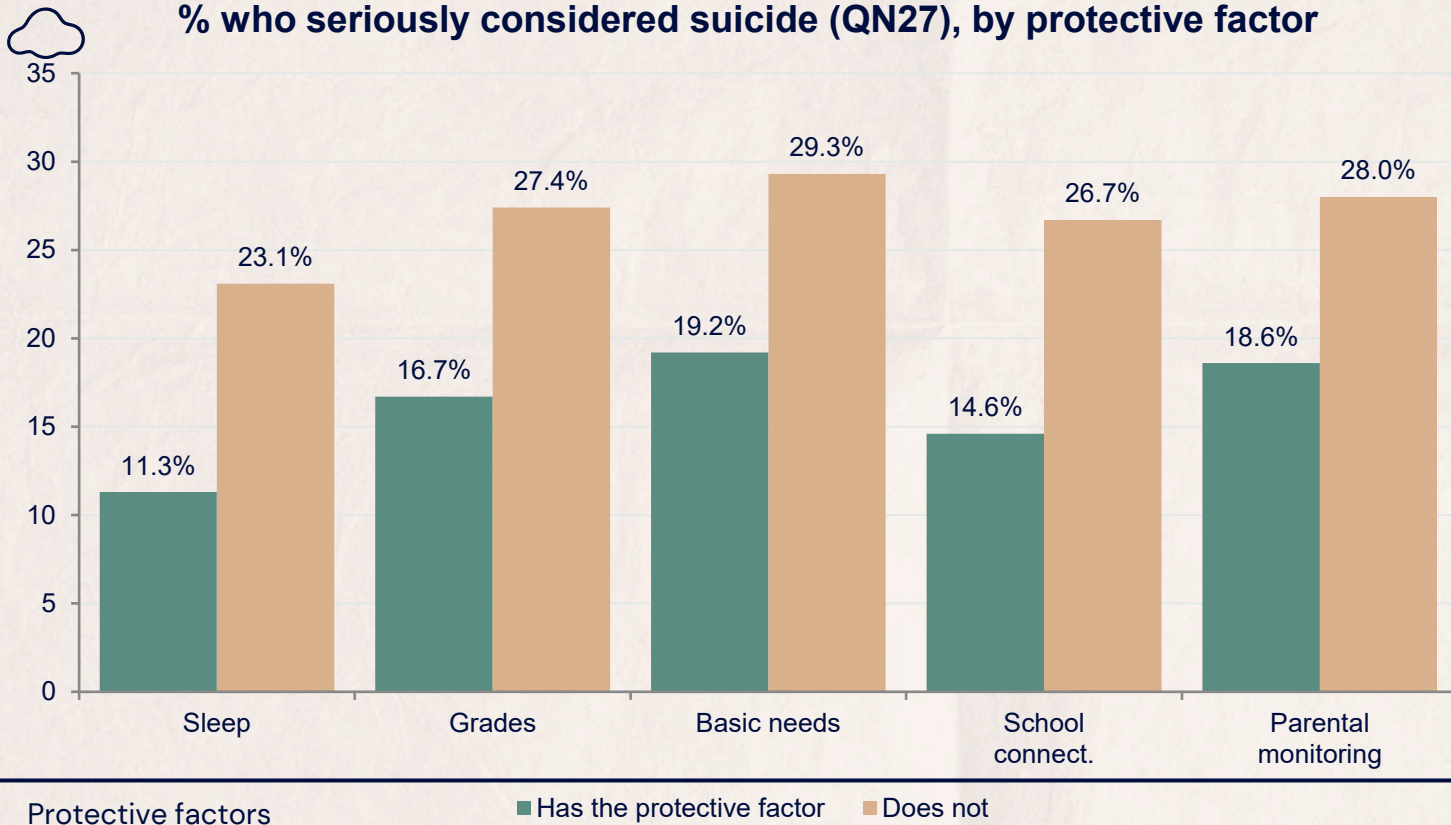
Suicidality Outcomes by Race/Ethnicity

Weighted % with 95% CI, all four outcomes



Aliya's variables

% who seriously considered suicide (QN27), by protective factor



**HOLDS ACROSS
ALL 20 COMPARISONS**

Every 1 of the 5 protective factors showed lower prevalence on **every one** of the 4 suicidality outcomes:

1. Considered
2. Planned
3. Attempted
4. Injured

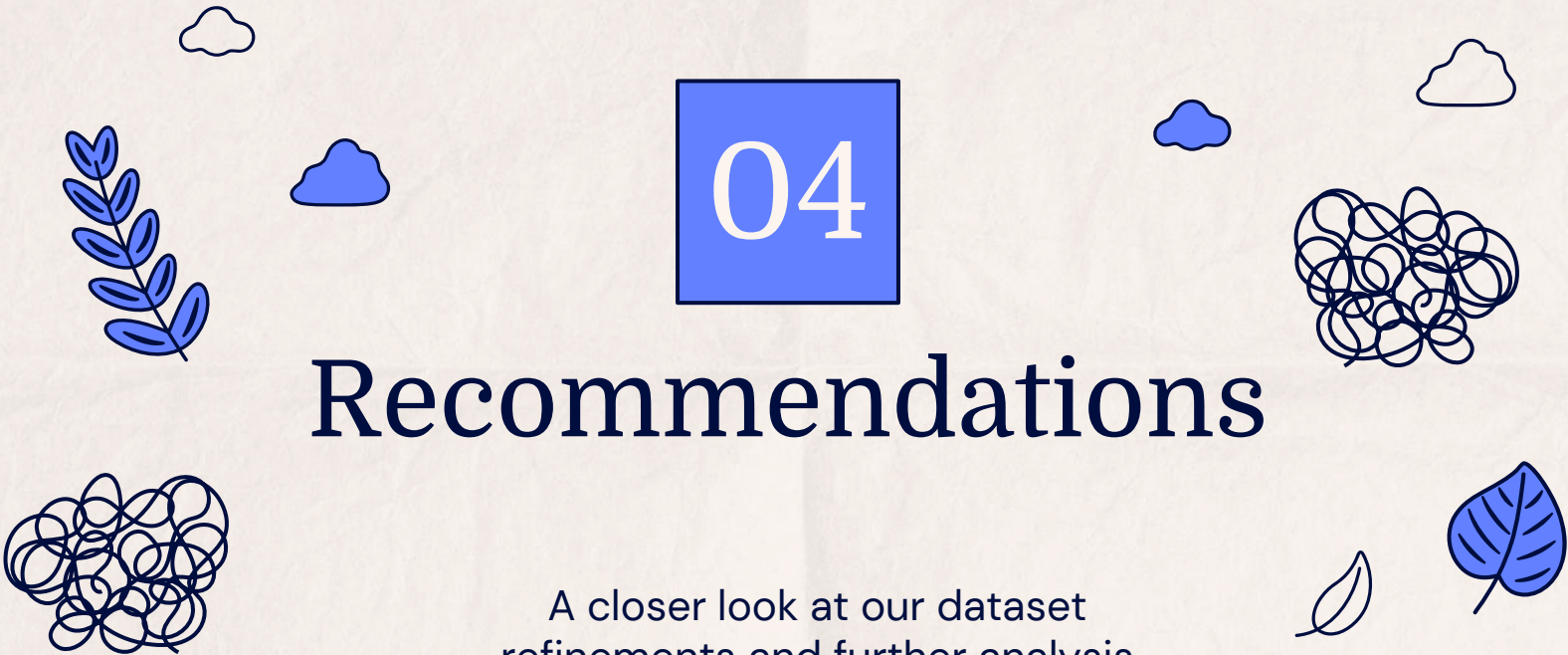
Aliya's variables (Protective factors)

Factor	CDC adjusted PR (2023 YRBS)	Our ratio (unadjusted)	Agreement
Sleep (QN85)	0.53 - 0.67	0.48 - 0.49	Close match
Basic needs met (QN99)	0.41 - 0.80	0.23 - 0.66	Close match
Parental monitoring (QN104)	0.51 - 0.74	0.34 - 0.66	Close match
School connectedness (QN103)	0.63 - 0.70	0.52 - 0.56	Close match
Grades (QN87)	Not tested by CDC	0.34 - 0.66	Consistent with outside literature

A ratio below 1.0 = lower risk in the protective-factor group. 0.53 $\approx$ roughly half the risk.

Sleep, basic needs, parental monitoring and school connectedness are the exact protective factors CDC selected for its own 2023 YRBS analysis: and our unadjusted ratios sit close to their adjusted ones.

Grades isn't one of CDC's six protective factors, so we drew on outside literature instead — directionally consistent, but the least consistently supported of the five across subgroups in the wider research.



04

Recommendations

A closer look at our dataset
refinements and further analysis



Limitations



Direction and causality

This is cross-sectional, self-report data. We can't establish that a protective factor came before (or caused a change in) an outcome, only that they co-occur



Recall & reference periods

Different questions ask about “during your life,” “past 12 months,” and “past 30 days”. These inconsistent windows can affect recall and comparability



Overlapping factors

Protective factors are interrelated (i.e., a well-monitored student may also sleep more), so this analysis can't isolate each factor's independent contribution



Unmeasured cofounders

Household socioeconomic status is known to shape both protective factors and mental health outcomes, however, isn't captured in the YRBS



05

Conclusions

A summary of our conclusions,
limitations, and next steps

Two sides of the same picture



Adverse experiences

- Sexual & dating violence,
- household dysfunction, school & community adversity, violence-adjacent behaviours each individually linked to higher suicide-risk indicators in prior research

↑ higher prevalence



Protective factors

- Sleep, grades, basic needs met, school connectedness, parental monitoring, each associated with substantially lower prevalence of every suicidality outcome measured.

↓ lower prevalence